Charpy Impact and Jominy End-Quench

Raw readings as recorded in the lab notebook

UMT312L Mechanical Testing Laboratory · Department of Materials Engineering, IISc
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Lab performed 30 October 2026 · Material: EN-8 · Sheet prepared 13 November 2026

This sheet is **only the raw data and the arithmetic that follows directly from it**. Nothing here is estimated, filled in or taken from a textbook. It is meant for checking the numbers against another group before the report is written. Every query raised in the first version of this sheet has now been settled against the notebook, the specimens and the course handout, and the answers are written into the tables.

1. Test conditions

Pendulum mass	21.79 kg
Pendulum capacity	**300 J**
Electronic (machine) reading error	± 2.26 J
Manual (dial) reading error	± 0.2 J
Cooling medium	liquid nitrogen bath (Friday Gp 1 is the liquid-N_2 group)
Vickers load	100 kgf, 10 s dwell *(fixed by the hardness handout)*
Rockwell scale for Jominy	C (as written at the head of the notebook page)
Jominy austenitising	**900 °C, 30 min** (handout, Friday Gp 1)

2. Charpy impact — as recorded

Two temperatures were written for every sample. The **initial** temperature is the reading taken as the specimen left the bath, i.e. the temperature of the test; the **final** temperature is the reading afterwards.

Sample	Condition	W (mm)	H (mm)	T_{initial}	T_{final}	E. energy	M. energy
Cu	copper	9.96	10.06	−80 °C	−20 °C	211.72 J	212 J
SS	stainless steel	9.92	9.94	−61 °C	−21 °C	299.86 J†	**107 J**
3	air cooled (AC)	9.92	9.94	−49 °C	−14 °C	4.59 J	3.0 J
4	water quenched (WQ)	9.99	9.99	−55 °C	−9 °C	4.00 J	2.3 J
5	furnace cooled (FC)	9.74	9.92	−63 °C	0 °C	3.41 J	2.0 J

Specific impact energy. The handout defines it as the absorbed energy divided by the effective cross-section under the notch, $A = Wb$, where the 2 mm notch removes 2 mm from the 10 mm dimension. Taking $A = (H - 2) \times W$:

Sample	$A = (H-2)W$ (mm^2)	a_k from E. energy (J/cm^2)	a_k from M. energy (J/cm^2)
Cu	80.28	263.73	264.08
SS	78.76	380.70†	**135.85**
3 (AC)	78.76	5.83	3.81
4 (WQ)	79.82	5.01	2.88
5 (FC)	77.14	4.42	2.59

The notch is cut across one of the larger (55 × 10 mm) faces, so it removes 2 mm from the dimension perpendicular to that face and the fracture area is $W \times (H - 2)$ (handout, section on the

specimen). Interchanging W and H would change every a_k by under 2 %, since the measured pairs differ by at most 0.18 mm.

Reported energies are the manual (dial) readings. The dial is the method the handout describes and its error is ±0.2 J against ±2.26 J for the electronic display.

Two readings that do not agree with each other:

- † Stainless steel: electronic 299.86 J against manual 107 J. The pendulum capacity is 300 J, so 299.86 J is the full swing to within 0.05 % — the display failed to register the loss rather than measuring one. **Use 107 J.**
- Sample 4: electronic 4.00 J against manual 2.3 J — a gap of 1.7 J, inside the ±2.26 J machine error.
- For Cu the two agree to 0.3 J (211.72 vs 212), so the machine is working on large energies.

3. Vickers hardness (100 kgf, 10 s)

Three indents per sample, two diagonals each, all in mm.

	indent 1		indent 2		indent 3		**mean** d	**HV$_{100}$**
Notebook table	d_1	d_2	d_1	d_2	d_1	d_2	(mm)	
①	1.028	1.034	1.033	1.035	1.032	1.038	1.0333	**173.7**
②	0.875	0.810	0.810	0.896	0.815	0.906	0.8520	**255.5**
③	1.112	1.134	1.104	1.129	1.109	1.132	1.1200	**147.8**

$HV = 1.8544\,F/d^2$ with $F = 100$ kgf and d in mm. The three mean diagonals were recomputed from the six readings and match the notebook exactly.

Which hardness belongs to which specimen. The three samples were indented in no fixed order, so the notebook's tables ①, ②, ③ are not in specimen order. The pairing does not depend on that order, because two independent records fix it:

1. The heat-treatment key written in the notebook: **3 = AC, 4 = WQ, 5 = FC** (reproduced in the appendix).
2. For one steel the order of hardness cannot reverse. Furnace cooling gives coarse pearlite (softest), air cooling gives fine pearlite (in between), water quenching gives martensite and bainite (hardest). Sorting the three measured hardnesses therefore assigns each one uniquely.

notebook table	mean d (mm)	HV_{100}	specimen and condition
②	0.8520	255.5	**sample 4** — water quenched
①	1.0333	173.7	**sample 3** — air cooled
③	1.1200	147.8	**sample 5** — furnace cooled

A third, independent check. The water-quenched sample reads 255.5 HV, about 23 HRC. The Jominy bar of the same steel reads 23 HRC at roughly 12 mm from the quenched end — the cooling rate a 10 mm square bar sees in its interior during a water quench. Two different experiments, the same answer.

4. Jominy end-quench — Rockwell C against distance

x (mm)	HRC	x (mm)	HRC	x (mm)	HRC
0	**50** (scatter 43–50)	9	24	25	17
1.5	27	10.5	22.25	30	16
3.0	26.5	12	23	35	15
4.5	26	13.5	21.5	40	14
6.0	25	15	21	45	13.5
7.5	23.75	20	19	50	13

Note on the 0 mm point: repeat measurements at the quenched face scattered over roughly 43 to 50 HRC. **50 HRC is reported.** The end face is scaled, decarburised and not ground square to the indenter — every one of those drives a Rockwell reading *down*, never up — so the scatter is one-sided and the highest reading is the best estimate. It is also the one closest to the ≈55–57 HRC expected of fully martensitic 0.4 wt% C steel. ASTM A255 starts its traverse at 1.6 mm, not at 0, so this point is excluded from any fit anyway. **Note on 15 mm:** the figure is between a 20 and a 21; read again on 13 Nov as **21**.

5. What this data does and does not establish

Established by the measurements themselves:

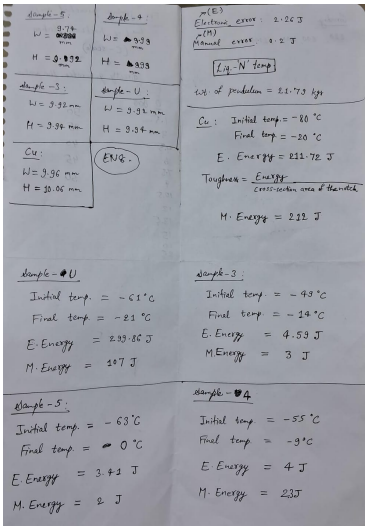
- The three EN-8 heat treatments differ clearly in hardness: 147.8, 173.7 and 255.5 HV.
- The hardest of the three is 255.5 HV ≈ 23 HRC. The Jominy bar of the same steel reads 23 HRC at about 12 mm from the quenched end — a cooling rate consistent with the interior of a 10 mm square bar quenched in water.
- Copper absorbed 212 J and the stainless steel 107 J at these temperatures, while all three EN-8 samples absorbed 2–3 J.
- The Jominy hardness falls steeply over the first 1.5 mm (50 to 27 HRC) and then decays slowly to 13 HRC at 50 mm.
- **Fracture surfaces** (photographs at the end of this sheet). All six EN-8 halves: flat and shiny, 100 % cleavage, no shear lips, no lateral expansion — the three heat treatments are indistinguishable by eye. Stainless steel: dull, fine and fibrous, chevron markings, shear lips. Copper: *did not break* — bent into a V with a deep waist at the notch, so its 212 J is a lower bound.

Not established, and not to be claimed:

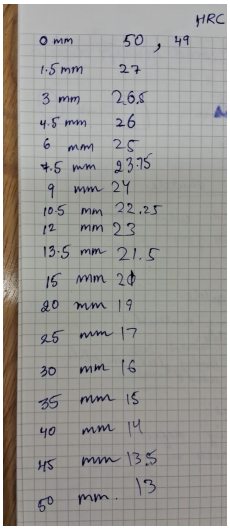
- The three EN-8 impact energies (3.0, 2.3, 2.0 J by dial) are *smaller than the ±2.26 J machine error*. They cannot be ranked against each other. They show only that all three conditions are fully brittle at these temperatures.
- Each sample was tested at one temperature only, and the three temperatures differ (−49, −55, −63 °C), so even the apparent ordering is confounded.
- No transition temperature, no 27 J index and no 50 % FATT can be obtained from single-temperature data. These need the other groups' results.
- Since the copper never separated, its 212 J is the work of bending and tearing, not a true Charpy fracture energy. Groups that broke their copper at a higher temperature should say so when the numbers are pooled.

Appendix — the original record, on one page

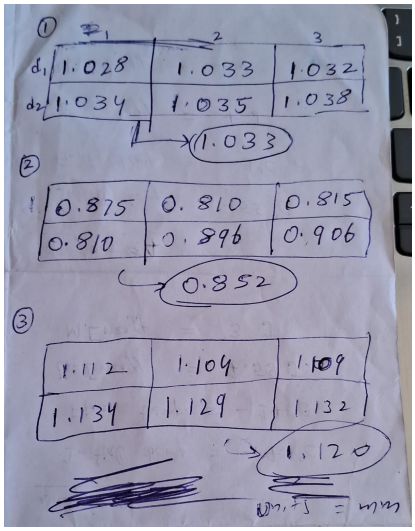
The notebook pages written in the lab on 30 October 2026, and the broken specimens photographed on 13 November 2026. Small, for reference — every number read off them is in the tables above.



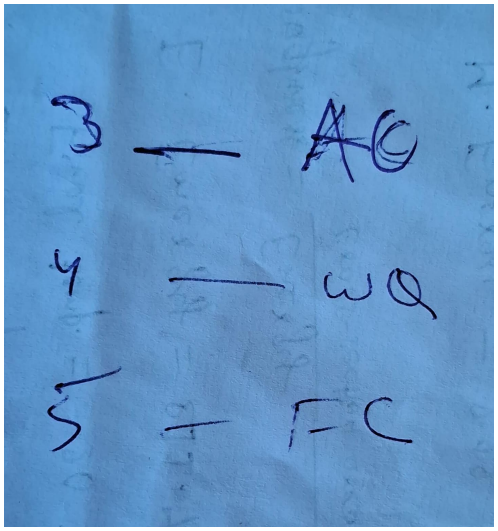
(a) Charpy: dimensions, machine errors, temperatures, energies.



(b) Jominy: Rockwell C against distance from the quenched end.



(c) Vickers: three indents per sample, two diagonals each, in mm.



(d) Heat-treatment key: 3 air cooled, 4 water quenched, 5 furnace cooled.



(e) All five after test. Left: the copper, bent into a V, still in one piece. Centre: the six EN-8 halves, flat and faceted. Bottom: the two stainless steel halves, dull and fibrous.



(f) Closer. The granular EN-8 faces are cleavage; the copper has torn, not cleaved; the stainless steel half at the right shows chevrons and a shear lip.